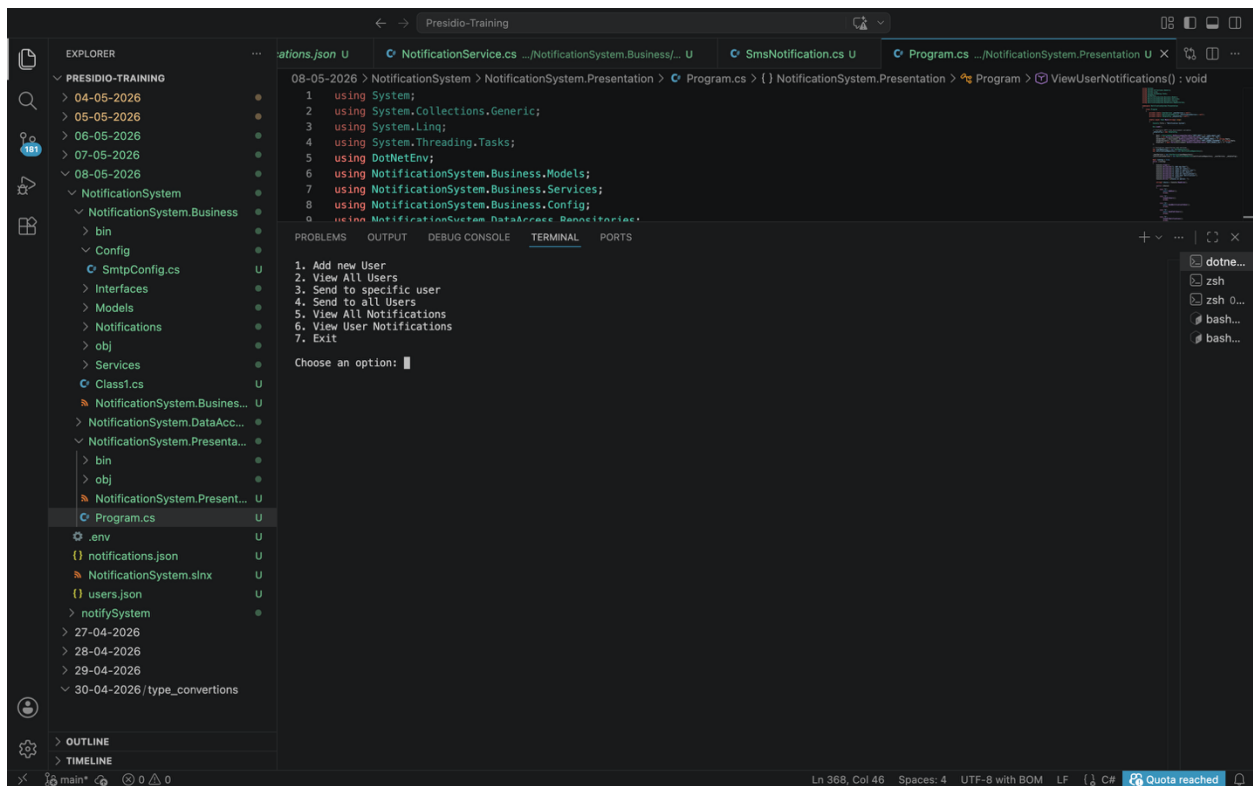


Assignment: Simple Notification System Using 3-Tier Architecture

Source code GitHub Url: <https://github.com/Padmavathi-SJ/Presidio-training/tree/main/08-05-2026/NotificationSystem>

OUTPUT:

Menu to choose operation:



The screenshot displays the Visual Studio Code interface for a project named "Presidio-Training". The Explorer pane on the left shows the project structure, including folders for dates (04-05-2026 to 30-04-2026) and a "NotificationSystem" folder. The "NotificationSystem" folder contains subfolders for "Business" and "Presentation", along with files like "bin", "Config", "SmtConfig.cs", "Interfaces", "Models", "Notifications", "obj", "Services", "Class1.cs", "NotificationSystem.Business...", "NotificationSystem.DataAcc...", "NotificationSystem.Presenta...", "Program.cs", ".env", "notifications.json", "NotificationSystem.slnx", "users.json", and "notifySystem". The "Program.cs" file is selected and open in the editor, showing the following code:

```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Threading.Tasks;
5 using DotNetEnv;
6 using NotificationSystem.Business.Models;
7 using NotificationSystem.Business.Services;
8 using NotificationSystem.Business.Config;
9 using NotificationSystem.DataAccess.Repository;
```

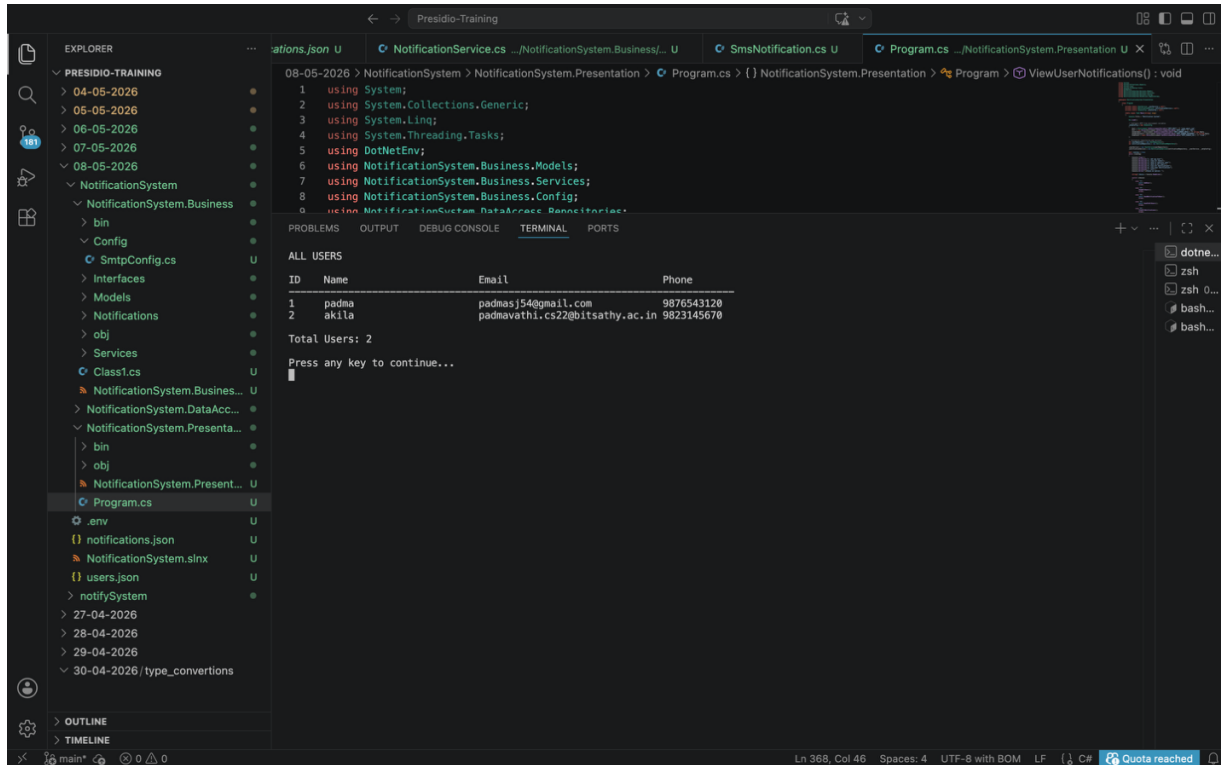
The Output pane at the bottom shows the program's execution, displaying a menu of options:

```
1. Add new User
2. View All Users
3. Send to specific user
4. Send to all Users
5. View All Notifications
6. View User Notifications
7. Exit

Choose an option: █
```

The status bar at the bottom indicates the current line and column (Ln 368, Col 46), the number of spaces (4), the encoding (UTF-8 with BOM), the line feed (LF), the character set (C#), and a "Quota reached" warning.

Display all active users:



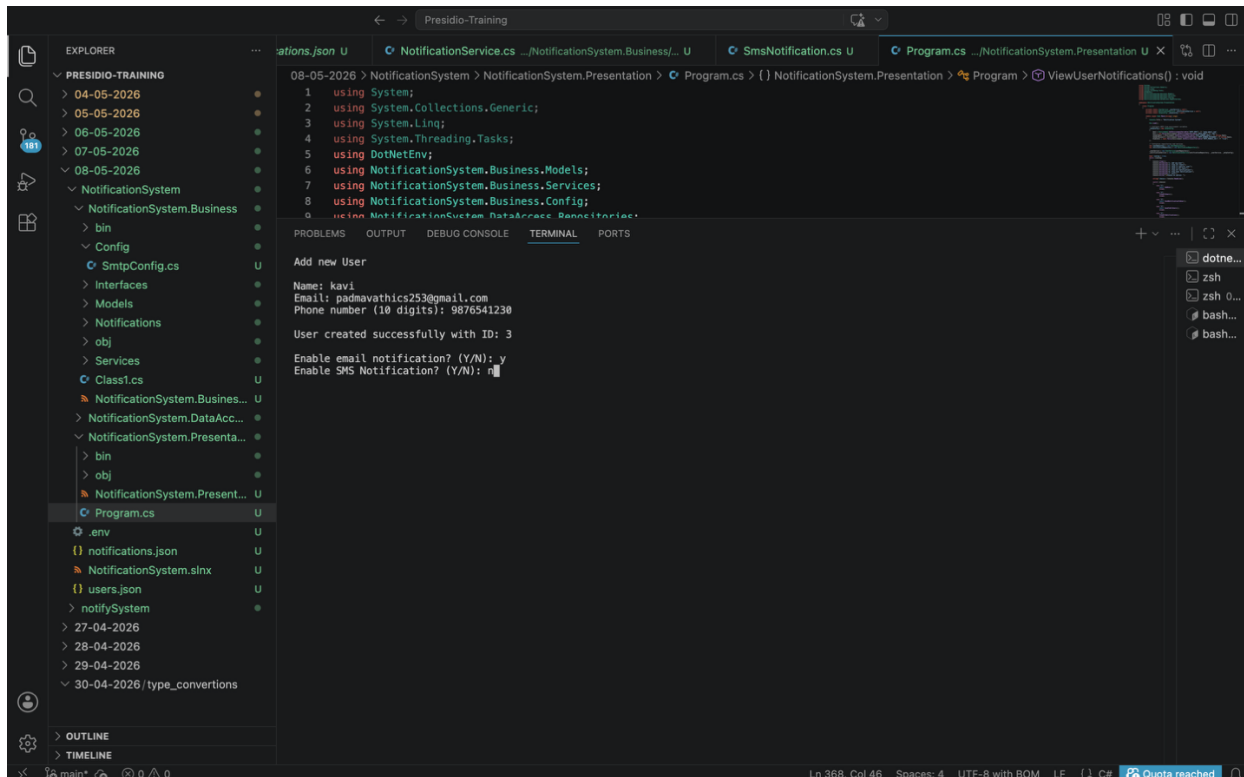
```
08-05-2026 > NotificationSystem > NotificationSystem.Presentation > Program.cs > {} NotificationSystem.Presentation > Program > ViewUserNotifications() : void
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Threading.Tasks;
5 using DotNetEnv;
6 using NotificationSystem.Business.Models;
7 using NotificationSystem.Business.Services;
8 using NotificationSystem.Business.Config;
9 using NotificationSystem.DataAccess.Repository;

ALL USERS
ID Name Email Phone
-----
1 padma padmas54@gmail.com 9876543128
2 akila padmavathi.cs22@bitsathy.ac.in 9823145678

Total Users: 2

Press any key to continue...
```

Add new user:



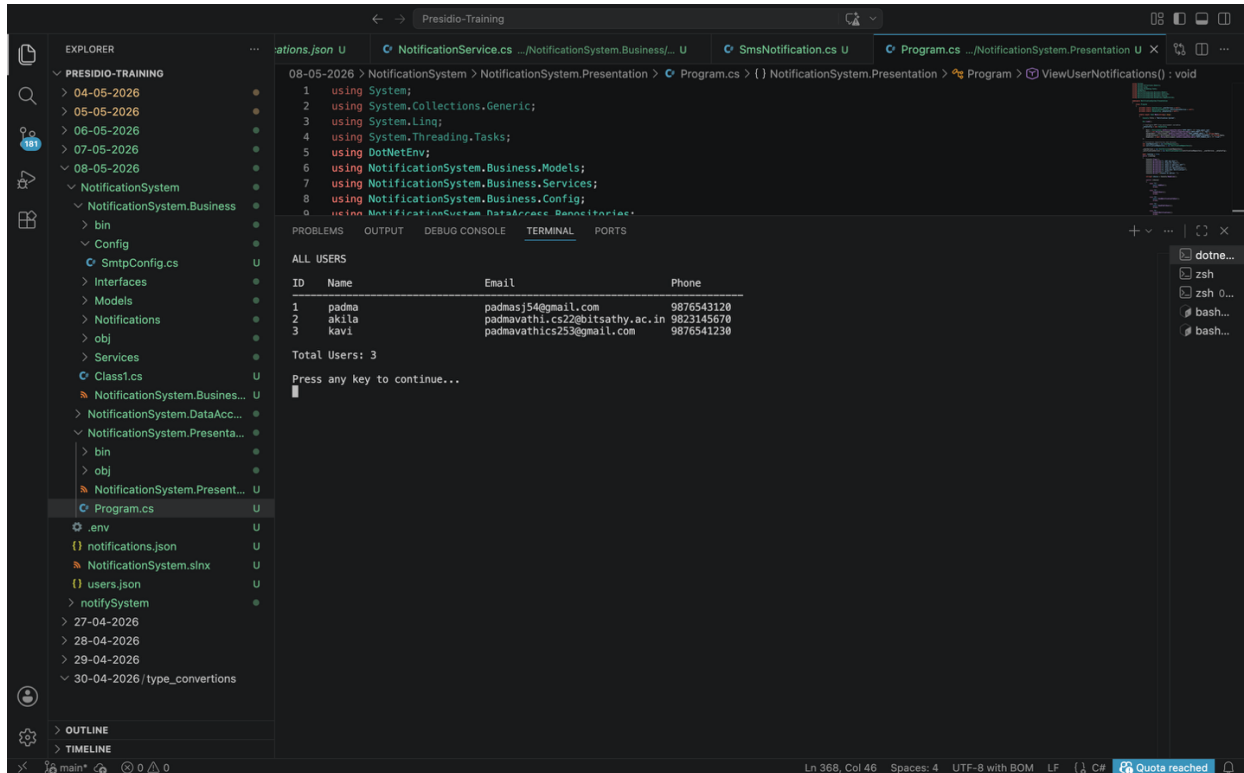
```
08-05-2026 > NotificationSystem > NotificationSystem.Presentation > Program.cs > {} NotificationSystem.Presentation > Program > ViewUserNotifications() : void
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Threading.Tasks;
5 using DotNetEnv;
6 using NotificationSystem.Business.Models;
7 using NotificationSystem.Business.Services;
8 using NotificationSystem.Business.Config;
9 using NotificationSystem.DataAccess.Repository;

Add new User
Name: kavi
Email: padmavathics253@gmail.com
Phone number (10 digits): 9876541230

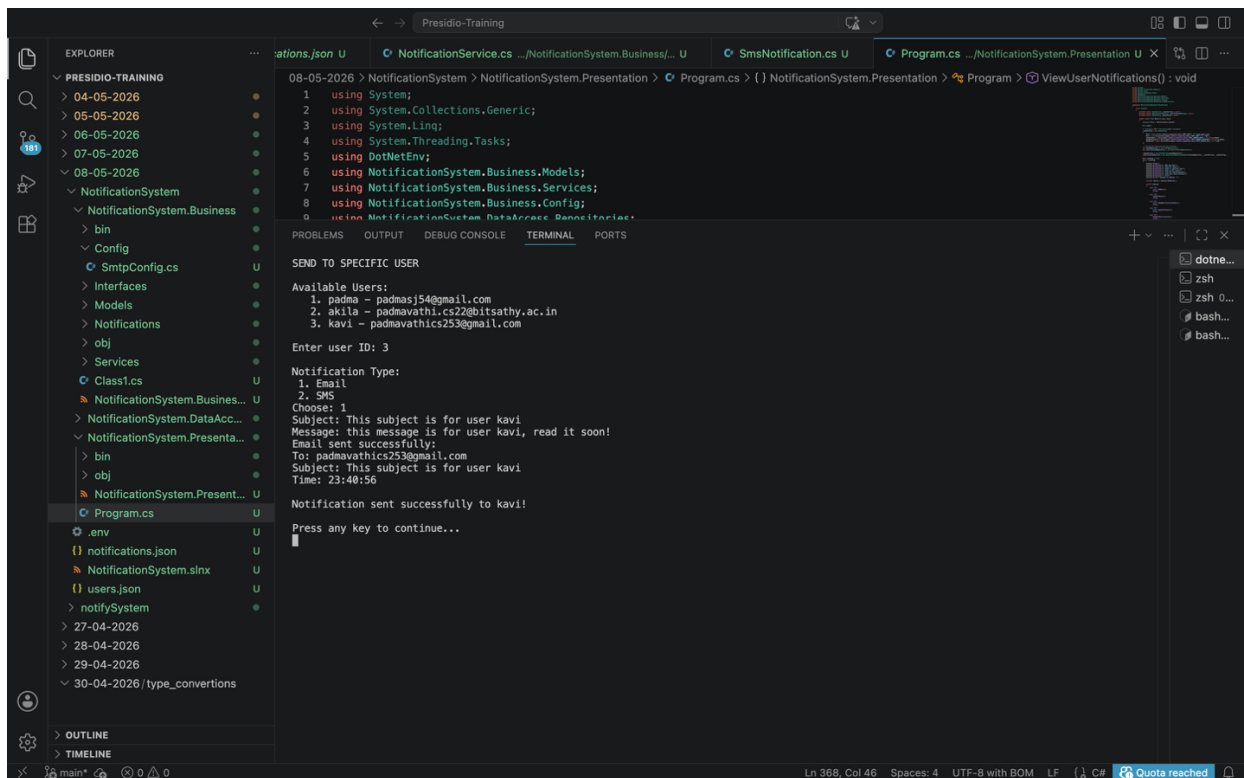
User created successfully with ID: 3

Enable email notification? (Y/N): y
Enable SMS Notification? (Y/N): n
```

Display all available users



Send email notification to a specific user:



Send email notification to all available users:

The screenshot shows a Visual Studio Code editor with a project named "Presidio-Training". The Explorer pane on the left shows the project structure, including folders for "NotificationSystem", "NotificationSystem.Business", and "NotificationSystem.Presentation". The Solution Explorer on the right shows the "Program.cs" file selected. The main editor area displays the code for "Program.cs" in the "NotificationSystem.Presentation" namespace. The code includes using statements for System, System.Collections.Generic, System.Linq, System.Threading.Tasks, and DotNetEnv, followed by using statements for NotificationSystem.Business.Models, NotificationSystem.Business.Services, and NotificationSystem.Business.Config. The main logic is in the "ViewUserNotifications()" method, which sends email notifications to all active users. The output window shows the execution results, including the message "SEND TO ALL USERS" and the details of the email notifications sent to three users: padmasj54@gmail.com, padmavathi.cs22@bitsathy.ac.in, and padmavathics253@gmail.com. The summary indicates that 3 emails were sent successfully and 0 failed.

```
08-05-2026 > NotificationSystem > NotificationSystem.Presentation > Program.cs > {} NotificationSystem.Presentation > Program > ViewUserNotifications() : void
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Threading.Tasks;
5 using DotNetEnv;
6 using NotificationSystem.Business.Models;
7 using NotificationSystem.Business.Services;
8 using NotificationSystem.Business.Config;
9 using NotificationSystem.DataAccess.Repository;

SEND TO ALL USERS
Will send to 3 active user(s).
Notification Type:
1. Email
2. SMS
Choose: 1
Subject: this subject for all 3 users!
Message: this message content is to reach you all 3 users soon!
Email sent successfully:
To: padmasj54@gmail.com
Subject: this subject for all 3 users!
Time: 23:41:50
padma.... Yes
Email sent successfully:
To: padmavathi.cs22@bitsathy.ac.in
Subject: this subject for all 3 users!
Time: 23:42:02
akila.... Yes
Email sent successfully:
To: padmavathics253@gmail.com
Subject: this subject for all 3 users!
Time: 23:42:06
kavi.... Yes
Summary: 3 sent, 0 failed
Press any key to continue...
```

Get and view all sent notifications:

The screenshot shows the same Visual Studio Code editor with the "Presidio-Training" project. The Explorer pane on the left shows the project structure, including folders for "NotificationSystem", "NotificationSystem.Business", and "NotificationSystem.Presentation". The Solution Explorer on the right shows the "Program.cs" file selected. The main editor area displays the code for "Program.cs" in the "NotificationSystem.Presentation" namespace. The code includes using statements for System, System.Collections.Generic, System.Linq, System.Threading.Tasks, and DotNetEnv, followed by using statements for NotificationSystem.Business.Models, NotificationSystem.Business.Services, and NotificationSystem.Business.Config. The main logic is in the "ViewUserNotifications()" method, which retrieves and displays all sent notifications. The output window shows the execution results, including the message "ALL NOTIFICATIONS" and a table of the notifications sent. The table has columns for ID, Type, To, Message, and Status. The notifications are listed in descending order of ID, with the most recent notification at the top. The status of each notification is "Sent" except for the last one, which is "SMTP SenderEmail not configured".

```
08-05-2026 > NotificationSystem > NotificationSystem.Presentation > Program.cs > {} NotificationSystem.Presentation > Program > ViewUserNotifications() : void
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Threading.Tasks;
5 using DotNetEnv;
6 using NotificationSystem.Business.Models;
7 using NotificationSystem.Business.Services;
8 using NotificationSystem.Business.Config;
9 using NotificationSystem.DataAccess.Repository;

ALL NOTIFICATIONS
ID Type To Message Status
8 Email padmavathics253@gmail.com this subject for all 3 users! Sent at 23:42
7 Email padmasj54@gmail.com this subject for all 3 users! Sent at 23:42
6 Email padmasj54@gmail.com this subject for all 3 users! Sent at 23:41
5 Email padmavathics253@gmail.com This subject is for user kavi Sent at 23:40
4 Email padmavathi.cs22@bitsathy.ac.in subject to 2 users Sent at 23:36
3 Email padmasj54@gmail.com subject to 2 users Sent at 23:36
2 Email padmasj54@gmail.com this is a subject Sent at 23:34
1 Email padmasj54@gmail.com this is subject 1 SMTP SenderEmail not configured
Press any key to continue...
```

Get and view all sent notifications of a user:

